



GEV Hazard Model — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

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1 Test Results — GEV Hazard Model (MKM-GH-001)

Tests run on **04 April 2026 at 14:48:33**.

Table 1: Test Results Summary — GEV Hazard Model (MKM-GH-001)

Total Tests	16
Passed	16
Failed	0
Skipped	0
Duration	0.32s

Table 2: Detailed Test Results — GEV Hazard Model

Test Description	Acceptance Criteria	Result	Time
Exceedance probability in range	<code>0.0 <= prob <= 1.0</code>	Pass	0.001s
Fit returns three floats	<code>isinstance(shape, float);</code> <code>isinstance(loc, float) (+1 more)</code>	Pass	0.018s
Force gumbel sets shape zero	<code>shape == 0.0</code>	Pass	0.001s
Higher threshold lower probability	<code>p_high < p_low</code>	Pass	0.000s
Return level increases with period	<code>r1_100 > r1_10</code>	Pass	0.000s
Compute all responses returns dict	<code>isinstance(responses,</code> <code>dict); len(responses) ==</code> <code>len(synthetic_flat_gauges) (+1</code> <code>more)</code>	Pass	0.000s
Compute response returns gauge re- sponse	<code>isinstance(response, GaugeResponse);</code> <code>response.gauge_id == gauge_id (+1</code> <code>more)</code>	Pass	0.000s
Response threshold flags	<code>response.exceeded.alert;</code> <code>response.exceeded.warning</code>	Pass	0.000s
Annual flood probabilities in range	<code>0.0 <= curve.annual_flood_prob.alert</code> <code><= 1.0; 0.0 <=</code> <code>curve.annual_flood_prob.warning</code> <code><= 1.0 (+1 more)</code>	Pass	0.097s
Build returns dict of curves	<code>isinstance(curves, dict); len(curves)</code> <code>== len(synthetic_flat_gauges) (+2</code> <code>more)</code>	Pass	0.095s
Each curve is gauge hazard curve	<code>isinstance(curve, GaugeHazardCurve);</code> <code>curve.gauge_id == gauge_id</code>	Pass	0.104s
Cumulative prob increasing	<code>probs[i] >= probs[i - 1]</code>	Pass	0.000s
Probabilities sum to one	<code>abs(point.survival_prob +</code> <code>point.prob.at.least.one - 1.0) ...</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Returns correct length	<code>len(ts) == 5</code>	Pass	0.000s
Returns term structure points	<code>isinstance(point, TermStructurePoint)</code>	Pass	0.000s
Survival prob decreasing	<code>survivals[i] <= survivals[i - 1]</code>	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
05-Mar-2026	1.0	Initial test suite (16 tests)	D. Kelly
04-Apr-2026	1.1	16 test(s) added; 16 test(s) removed (total: 32)	D. Kelly